Advancing the 3Rs (Reduce, Reuse, Recycle) through the Refurbishment of Medical Aids

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1. Introduction

The concept of advancing the 3Rs (Reduce, Reuse, Recycle) through the refurbishment of medical aids was conceptualised during the rotation at *Pusat Rehabilitasi PERKESO Tun Abdul Razak*. During the exposure to the supply process of medical aids, such as wheelchairs, to *Orang Berinsurans (OBs)*, it was observed that no structured mechanism currently exists for retrieving equipment once it is no longer in use. For instance, in cases where an OB requires a wheelchair temporarily due to a fractured leg and subsequently recovers, the equipment often remains unused without a formal process for return or redistribution, potentially contributing to avoidable waste generation.

The absence of a retrieval framework presents significant implications for PERKESO's Environmental, Social and Governance (ESG) footprint, particularly in relation to medical waste generation and inefficient resource utilisation.

In addition, with the expansion of protection coverage of *Skim Kemalangan Bukan Bencana Kerja (SKBBK)* which will be providing 24-hour coverage including for foreign workers, the projected volume of claims and rehabilitation cases is expected to increase by 3 times according to the Actuarial department. This expanded coverage will inevitably increase the demand for medical aids by 3 times as well.

International benchmarking indicates that circular rehabilitation models are both feasible and sustainable. A notable example is the Accident Compensation Corporation (ACC) of New Zealand. Under the ACC model, rehabilitation equipment remains part of the ACC-funded system rather than being permanently owned by patients in most cases. After use, the equipment is systematically retrieved, refurbished through controlled sterilisation and maintenance processes and redistributed to other eligible users following safety validation. In contrast, PERKESO's current framework operates largely as a "supply-and-forget" model.

2. Background: R-Pemulihan Project from Segamat Local Office

During the rotation in the Audit Team, it was highlighted by *pegawai tadbir*, Pn Nurulhuda Binti Yaakub, that PERKESO Segamat had previously initiated a similar project known as *R-Pemulihan* in 2017. Consultation was subsequently conducted with PERKESO Segamat *Pegawai Eksekutif Kanan*, En Mohd Syahmi Bin Samsuddin, who pioneered *R-Pemulihan*. It was confirmed that the project did not achieve full implementation due to structural and governance constraints.

Figure 1: Virtual Session with Mr. Mohd Syahmi from PP Segamat

2.1 Identified Constraints

a) **Absence of Official Directives:** A major barrier was identified in the absence of formal circulars *(pekeliling)* or gazetted rules explicitly authorising the retrieval of medical aids after issuance. Without written procedural authority, implementation resistance was observed across regional offices.

b) **Audit Concerns:** Internal auditors raised concerns regarding the storage of medical aids within PERKESO premises when such items were no longer recorded as PERKESO assets. The absence of clear classification between "issued assets" and "recoverable assets" resulted in accounting ambiguity.

c) **Storage and Logistics Limitations:** Temporary storage at Segamat Hospital was no longer available. Transportation assistance had previously been provided by Segamat Local Office. However, resistance was observed due to concerns relating to manpower allocation and logistical costs.

d) **Managerial Resistance:** Although conceptual support was shown by senior management, reluctance was observed among middle management and veteran officers in the absence of formalised Standard Operating Procedures (SOPs) and written directives.

3. Sessions with Putrajaya Local Office

Pn Norliza Binti Noordin, *Pengurus Pejabat PERKESO Putrajaya*, has previously worked closely with the *R-Pemulihan* initiative. She was therefore appointed as the supervisor of the present "Advancement of 3Rs and Refurbishment of Medical Aids" project.

With the assistance of case managers (CM) and *Skuad Sinergi* from the Putrajaya Local Office, an improved framework has been developed which will serve as the structural backbone of the entire project as seen below:

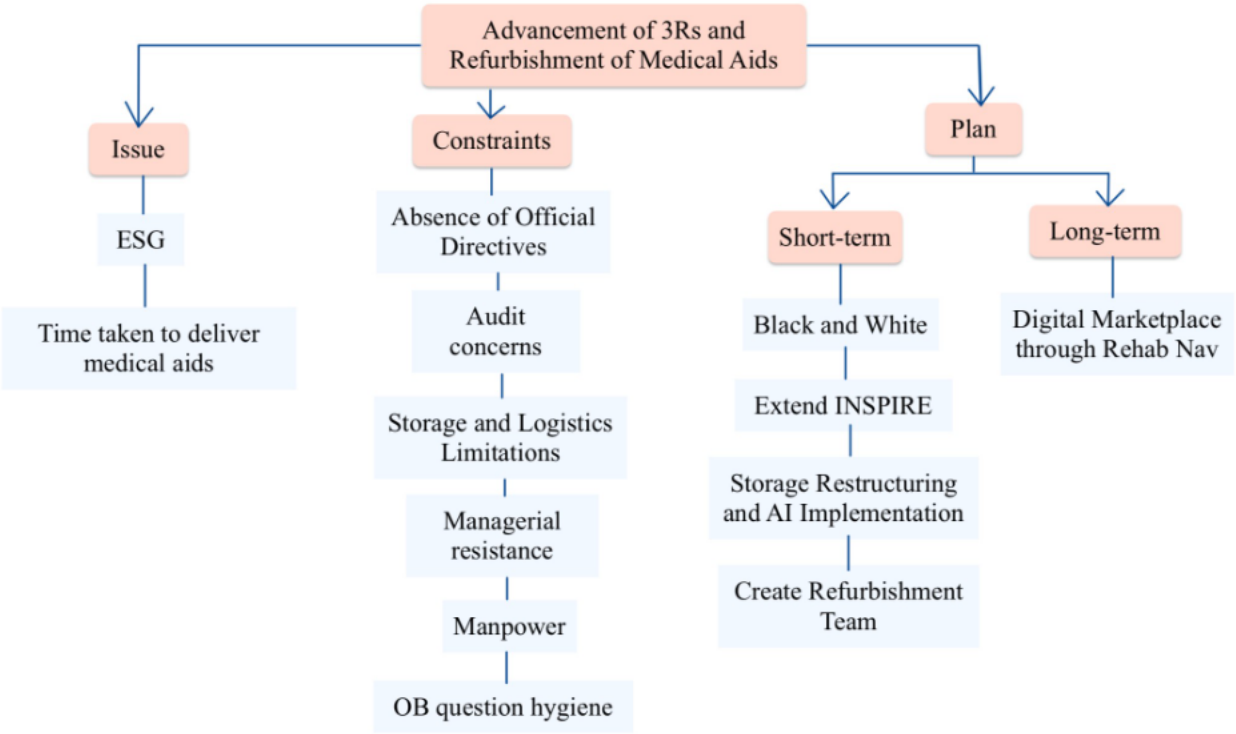

Figure 2: Capstone Project Framework

The officers involved are Pn Meilin Lampisung *(Pegawai Pengurusan Hilang Upaya)*, Pn Noor Syazwani Binti. Azmi *(Penolong Pegawai Pengurusan Hilang Upaya)*, En Ahmad Sapawie Bin Rokman *(Pegawai Eksekutif)*, Pn Mastura Bt Omar *(Pegawai Eksekutif Kanan)*, En Marwan Bin. Ibrahim *(Pegawai Eksekutif),* and En Mohd Ashreeq Amierullah Bin. Mohd Firdaus *(Pegawai Eksekutif)*.

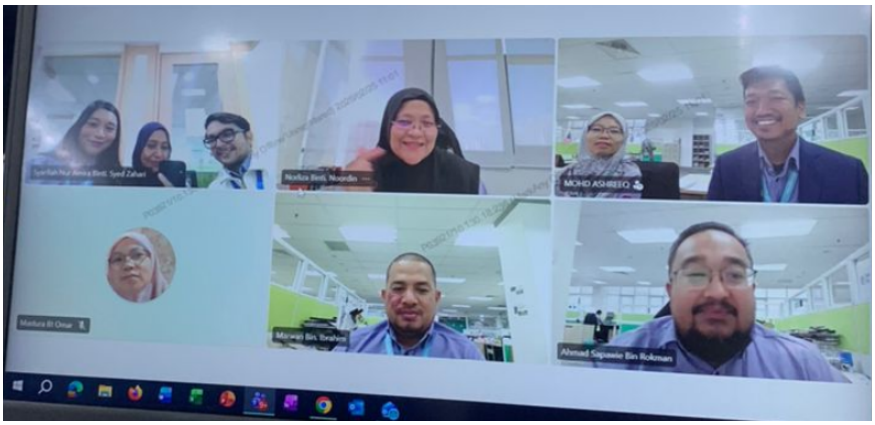

Figure 3: First Virtual Meeting with PP Putrajaya

Figure 4: Second In-person Meeting at Putrajaya Local Office

4. Proposed Solutions

Although the initiative is not entirely new, further research has been conducted and additional planning has been undertaken to ensure that this project is feasible for implementation under current conditions and remains sustainable, strong and relevant in the future.

4.1 Short-term Plans (Month 0–24)

4.1.1 Phase 1 (Month 0–3): Black and White

Under the current legislation, there is no requirement for medical equipment to be new, nor is there a regulatory provision explicitly allowing the retrieval of unused equipment. Additionally, there is no governance framework guiding the implementation of refurbished medical aids.

Proposed Actions to Operationalise Regulatory Flexibility

To formalise and enforce this initiative, the following actions will be implemented:

1. Introduction of a new regulatory clause permitting retrieval of unused medical aids under Section 57.
2. Issuance of a formal circular *(pekeliling)* to standardise nationwide implementation.
3. Legal review and compliance validation to ensure alignment with Act 4 and existing subsidiary regulations.
4. Formal asset reclassification framework distinguishing:
    - Permanently Issued Medical Aid
    - Conditionally Issued (Recoverable) Medical Aid

a) **Expected Outputs**

- Official regulatory directive ("black and white" authority).
- Approved governance framework supported by Standard Operating Procedures (SOPs).
- Clear asset lifecycle classification.
- Enforceable nationwide compliance structure.

With written authority in place, structured compliance across all Local Offices can reasonably be expected.

b) **Suggested Draft Clauses (For Board Consideration)**

- **Proposed Clause 1: Conditional Issuance**

Medical aids provided under Section 57 shall remain under the ownership of the Organization unless otherwise determined. Such medical aids may be issued on a conditional usage basis for the duration of medical necessity.

- **Proposed Clause 2: Right of Retrieval**

The Organisation reserves the right to retrieve medical aids issued under Section 57 upon cessation of medical necessity, recovery, permanent transition to alternative aids, case closure, or death of the insured person.

- **Proposed Clause 3: Refurbishment and Reallocation**

Retrieved medical aids may be refurbished, certified for safety and functionality, and reassigned to other eligible insured persons in accordance with approved standards and procedures.

- **Proposed Clause 4: Asset Classification**

Medical aids shall be classified as either:

(a) Permanently Issued Medical Aid; or

(b) Conditionally Issued (Recoverable) Medical Aid,

as determined by the Organisation based on clinical and operational criteria.

These clauses are:

- Legally defensible under Section 57(2) ("on such conditions as may be specified")
- Audit-friendly
- ESG-aligned
- Operationally enforceable

4.1.2 Phase 2 (Month 3–8): Extend INSPIRE

a) **Current INSPIRE System**
(Accessible only by Case Managers and PERKESO Doctors)

1. Medical aid orders for *Orang Berinsurans (OBs)*, such as wheelchairs, are submitted into the system by case managers at Local Offices based on hospital doctors' reports.
2. The requests are then assessed by PERKESO doctors from *BPPP (Bahagian Pencegahan, Perubatan, Pemulihan)*, and approvals are granted where appropriate.
3. Once approval has been obtained, the orders are processed by the case managers.

b) **Workflow – Case Manager (Local Office***)*

1. Different categories of cases, including ILAT, HUS, and HUK, are received by Local Office case managers (CMs).
2. An initial assessment is conducted through review of diagnosis and medical history.

3. Workplace visits are carried out, and determinations are made as to whether the rehabilitation goal is Return-to-Work (RTW) or Activity of Daily Living (ADL).
4. Job modification, job capacity evaluation, and work ability assessments are performed.
5. Monitoring is conducted for approximately 1–3 months to assess sustainability of the OB at work.
6. Case management is typically maintained for up to one year until case closure.
7. If rehabilitation intervention is required, the OB is referred to the rehabilitation team by the local office case manager.

c) Identified Monitoring Gap

At present, monitoring of medical aid usage is conducted primarily through the PHU *(Pemulihan Hilang Upaya)* Local Office and rehabilitation case managers. However, monitoring mechanisms for *Orang Berinsurans (OBs)* who have fully recovered or who have passed away are not systematically integrated into the INSPIRE system.

For PHU Local Office case managers, OBs who participate in PHU programme are closely monitored. However, the usability of medical aids is not monitored for OBs who do not participate in the PHU programme as medical aids are issued without structured follow-up of healing progression, effectively resulting in a supply-and-forget model without lifecycle tracking.

According to Local Office Putrajaya, non-PHU cases constitute most cases managed. Since non-PHU cases are higher, it creates even more material risk of medical aid underutilisation, premature replacement and avoidable procurement expenditure due to the absence of a structured lifecycle monitoring framework.

d) Proposed INSPIRE Prototype Extension

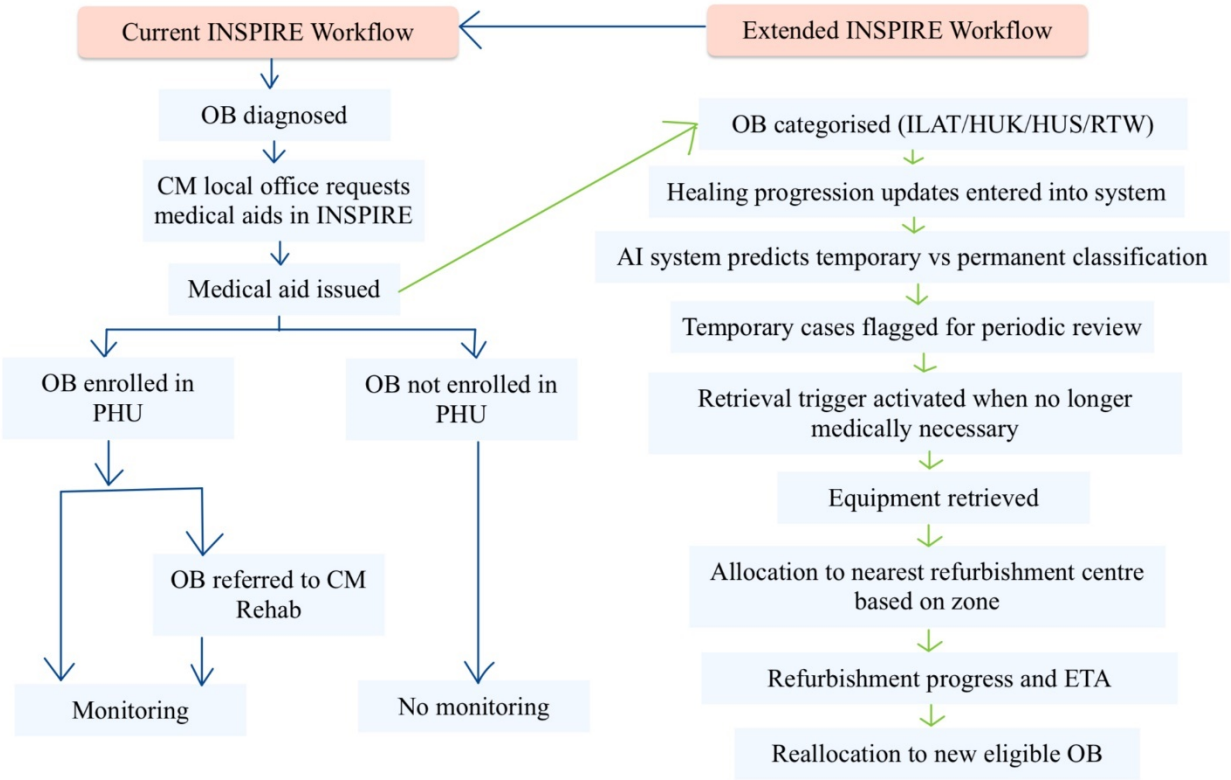

Figure 5: Extended INSPIRE Workflow

Under the proposed extension:

1. Data on OB progression will be incorporated into INSPIRE to enable early identification of medical aids that may be eligible for retrieval.
2. Updates on OB progression will be entered by case managers through a newly developed system function.
3. OBs will be categorised into structured groups such as ILAT, HUK, HUS and RTW to allow systematic monitoring and predictive tracking.
4. Based on historical case data and predictive modelling, early determination will be made as to whether a medical aid is temporary or permanent in nature.

5. Temporary cases will be automatically flagged for periodic review while permanent cases may be subjected to reduced monitoring frequency, with structured verification intervals established where appropriate.
6. A monthly digital verification mechanism may be incorporated, whereby case status (active, recovered, deceased, or transitioned) is updated within the system.
7. Predictions regarding the expected duration of medical aid usage will be generated based on diagnosis patterns and historical trends.

For example, for a stroke patient, initial wheelchair usage may be required, followed by transition to a walking frame as mobility improves. The wheelchair may be retrieved prior to issuance of the walking frame, optimising resource allocation.

The prototype for the extended INSPIRE system can be seen below and the QR code can be scanned to direct users to this.

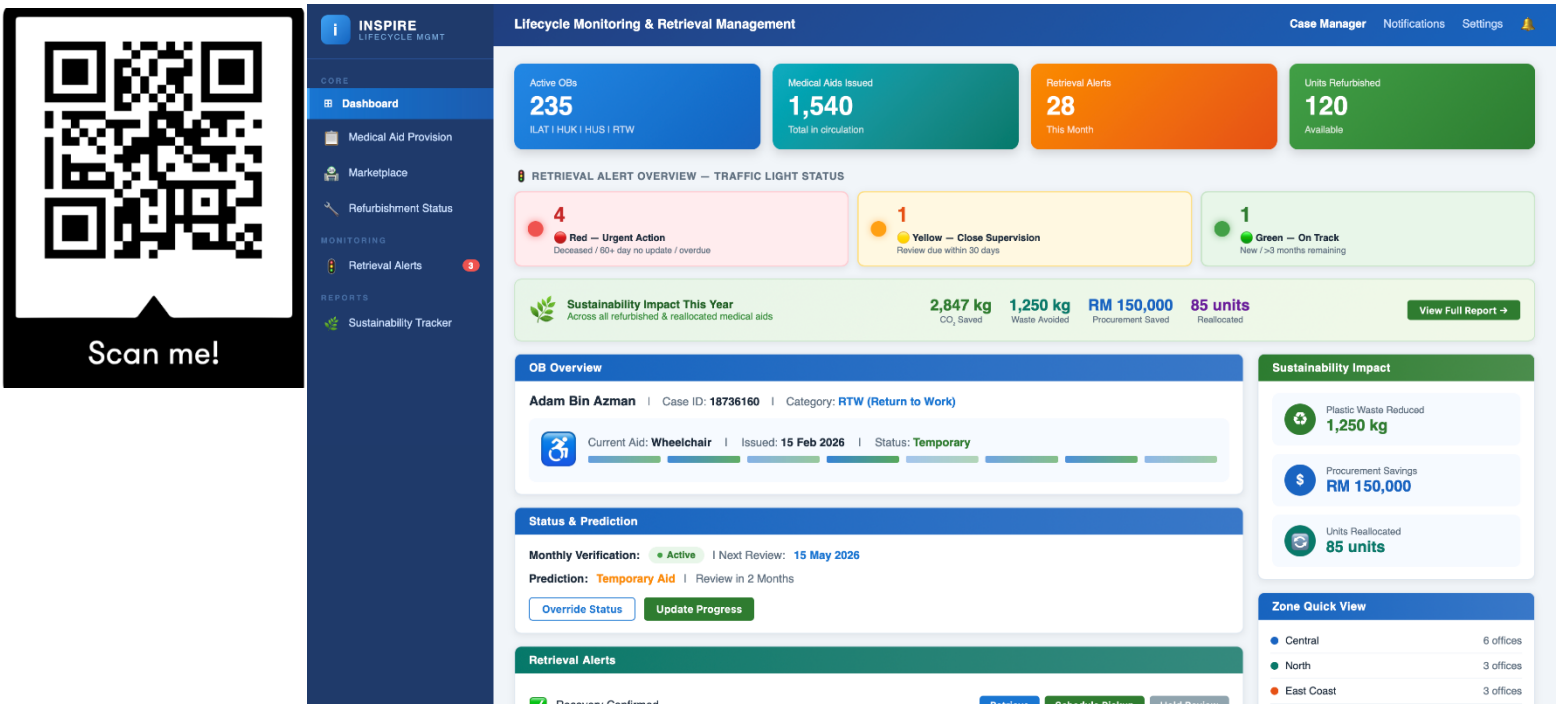

Figure 6: INSPIRE Prototype

4.1.3 Phase 3 (Month 8-14): Zone-based Refurbishment Centres

During the rotation at *Pusat Rehabilitasi PERKESO Tun Abdul Razak*, it was further noticed that:

- Individuals managing storage are also doing self-initiative for refurbishment of medical aids. Although there is a lack of manpower.
- Currently these refurbished medical aids are only used in Rehab Melaka.
- No dedicated team is responsible for lifecycle management of medical aids.
- With the implementation of SKBBK and PERKESO's overall projected benefits cases processing to be increased by 3x (1x of SBK, 2x of SKBBK), there is a need for a more efficient storage process for timeliness of process.

Figure 7: Pictures of the Storage Unit at at Pusat Rehabilitasi PERKESO Tun Abdul Razak

Proposed Framework:

a) **Refurbishment Team Creation (New Structured Unit)**

A dedicated Medical Aid Refurbishment Unit will be formally established to manage the refurbishment process effectively. The team will consist of:

- *Medical equipment technicians:* responsible for repairing and maintaining devices, sterilization officers to ensure all equipment meets hygiene standards, and safety compliance officers to oversee adherence to safety regulations.
- *Inventory control analysts:* will manage and monitor stock levels, logistics coordinators will handle the transportation and distribution of medical aids
- *Quality assurance personnel:* will ensure that all refurbished equipment meets the required quality standards before deployment.

b) Hygiene Assurance Framework

During informal engagements and feedback sessions with Orang Berinsurans (OBs), concerns were raised regarding the hygiene and safety of refurbished medical equipment. To address these concerns, PERKESO will implement a hygiene assurance framework to ensure that all refurbished equipment meets strict cleanliness and safety standards before redistribution. This will include industrial-grade cleaning procedures, advanced UV sterilisation technology, disinfection chambers, component-level sanitisation and contamination risk assessments.

Prior to redistribution, all refurbished equipment must undergo mandatory testing and certification. This includes safety inspections, functional performance testing, mechanical integrity verification and load-bearing validation. Electrical safety testing will also be conducted where applicable. Once approved, the equipment will receive quality assurance sign-off, certification tagging and digital re-registration in the INSPIRE system. No equipment will be redistributed without formal clearance.

c) Workforce Impact

This phase is expected to create new skilled roles within PERKESO while strengthening internal technical capabilities in medical aid refurbishment. The initiative will also reduce dependency on external procurement cycles and enhance institutional expertise in refurbishment processes.

Based on data from the Labour Market Exchange (LMX) website as of 6 March, the number of jobseekers recorded in Selangor was 11,435, followed by 4,591 in Kuala Lumpur and 434

in Putrajaya. The establishment of refurbishment-related roles therefore supports broader national workforce development objectives by contributing to employment opportunities within the region.

4.2 Long-term Plan (Month 24-36)

4.2.1 Phase 4 (Month 24–36): Digital Marketplace through Rehab Nav

From the data from the restructuring of storage units and the extended INSPIRE system, a Medical Aid Marketplace module will be integrated within Rehab Nav. It represents the demand–supply digital interface layer of the circular rehabilitation infrastructure. The Digital Marketplace prototype can be seen below:

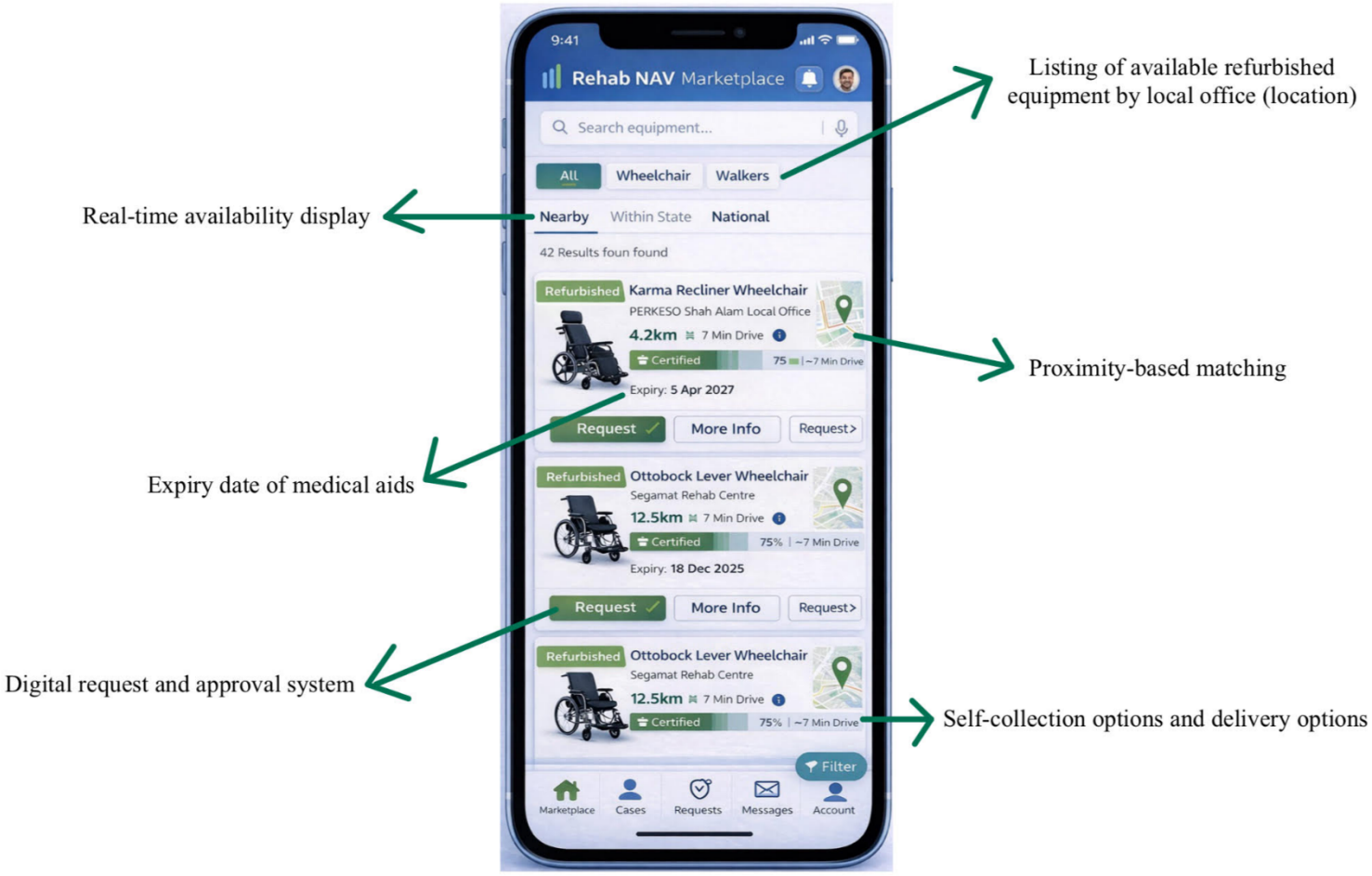

Figure 8: Digital Marketplace Prototype

Listing of refurbished equipment by local office / location

Ensures transparent visibility of available medical aids across the national network.

Proximity-based matching

Matches demand requests with the nearest available refurbished unit to reduce transport lag and optimise redistribution efficiency.

Digital request and approval system

Maintains governance control, enabling structured approvals while preserving compliance and audit trails.

Real-time availability display

Synchronised with the refurbishment team updates, ensuring up-to-date inventory visibility.

Delivery or self-collection options

Introduces operational flexibility while maintaining tracking continuity.

This functionality is implemented within the Rehab Nav application rather than INSPIRE, as INSPIRE already hosts multiple operational modules, while a mobile-based platform enables case managers to access the marketplace quickly during urgent medical aid requests.